CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 472 Practice

Midterm 472 · from L1 to L7

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01 About this midterm

This is a retyped copy of the 472 midterm, marked "From L1 to L7". The questions are copied as they are. It shows no mark distribution.

Important

I have no official answer key for this midterm. The solutions are my own, and every table was computed by code. The matrix chain, LCS and B-tree questions come from lectures whose slides I have not received, so I solved them with the standard textbook method. If the instructor's slides do it differently, follow the slides.

02 Questions

Q1 MATRIX CHAIN

Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABCD where the order for matrix:

A = 4 × 5 B = 5 × 8 C = 8 × 3 D = 3 × 7

Minimum operations = _____ Best order of multiplication: _____

	A	B	C	D
A	0	160		
B		0	120	
C			0	
D				0

Q2 TRUE / FALSE + MCQ

1. The code length in Huffman codes depend on the frequency of occurrence of characters. ○ True ○ False
2. Greedy algorithm can solve all the dynamic programming problems. ○ True ○ False
3. The time to run MERGESORT on a list of size n, T(n) = T(n/2) + T(n/2) + T(MERGE(n/2)) ○ True ○ False
4. In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size. ○ True ○ False
5. Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment. ○ True ○ False
6. An algorithm which does NOT use the past results and uses them to find new results is:

a. Greedy Method b. Divide and conquer c. Dynamic programming d. None of the mentioned

Q3 LCS

Using the dynamic programming approach, find the longest common subsequence between two strings:

X = ACBAED and Y = ABCADF.

Length _____ The subsequence: 1 _____ 2 _____

	⌀	A	B	C	A	D	F
⌀							
A							
C							
B							
A							
E							
D							

Q4 HASH TABLE

Insert integer keys A = {4, 13, 49, 6, 40, 33} in order into a hash table of size 9 using the hash function h1(k) = k mod 9. Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted.

0	1	2	3	4	5	6	7	8

Q5 B-TREE

A) Create a B-tree with order 5, min degree 2 the following item [24, 52, 90, 40, 11, 86, 61, 50]

B) insert 29

C) delete 52, 11